

Voter Model on Finite Graphs

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1 Introduction:

1.1 What Is Interacting Particular System (IPS)?

Consider a finite graph $G = (V, E)$ where V is a set of vertices and E is a set of edges. An IPS on G is a continuous-time Markov process $\zeta = (\zeta_t)_{t \geq 0}$ with state-space $S = A^V$ (i.e; state-space is all possible configurations where vertices are assigned values from set A), for some finite or countable set A and here $\zeta_t = \{\zeta_t(x) | x \in V\}$ denotes the configuration we got at time t . Thus $\zeta_t(x) = a, a \in A$ denotes that the system at vertex x at time t equals to state a .

In the most common practice we take $A = \{0, 1\}$, i.e; using 2 possible values which we may interpret as that $\zeta_t(x) = 1$ means there is a "particle" at site x and $\zeta_t(x) = 0$ means the site x is empty.

1.2 Flip Rates:

Assuming $A = \{0, 1\}$, a flip rate is a function $c : (V \times S) \rightarrow \mathbb{R}^+$, where $c(x, \eta)$ is the transition rate at which the state at site x changes in the configuration η . Equivalently we can say, at this rate the IPS evolves from a given $\eta \in S$ to the configuration $\eta^x \in S$ obtained from η by changing the state at site x , i.e; $\eta \rightarrow \eta^x$ at rate $c(x, \eta)$ with- $\eta^x(y) = \begin{cases} \eta(y), & \text{if } y \in V \setminus \{x\} \\ (1 - \eta(x)) & \text{if } y = x. \end{cases}$

If this rate $c(x, \eta)$ depends on η only via $\eta(x)$ (the value of the spin at x) then ζ consists of independent spin-flips. In general, this rate to change the spin at x depends on the neighbourhood of x . This dependence models an interaction between the spins at different sites. In order for ζ to be well-defined, the following restrictions must be placed on the local transition-rates $c(x, \eta)$:

- (i) $c(x, \eta)$ must depend on the states at far away sites.
- (ii) $c(x, \eta)$ must not be too large, i.e; bounded away from infinity.

1.3 Shift-Invariant Spin-Flips:

A spin-system is called shift-invariant if the rules of the dynamics are the same everywhere in the lattice, i.e; $c(x, \eta) = c(x + y, \tau_y \eta) \quad \forall y \in \mathbb{Z}^d$ where $\tau_y \eta(x) = \eta(x - y), \forall x \in \mathbb{Z}^d$ which means that flip

rate at site x depends on the configuration around x in exactly the same way it depends on the configuration around the origin.

1.4 Attractive Spin-Flips:

A spin-flip system is called attractive if the dynamics preserve the natural partial order on configurations. Define partial order $\eta \preceq \zeta$ if $\forall x, \eta(x) \leq \zeta(x)$ on the space of possible configurations. Now system is attractive if two processes η_t and ζ_t start with initial configurations $\eta_0 \preceq \zeta_0$, then we can couple them such that $\eta_t \preceq \zeta_t, \forall t \geq 0$. This guarantees monotonicity of the system.

Now these properties shift-invariance and attractiveness allow some interesting conclusions of the systems concerning their equilibrium and convergence to equilibrium. Some examples of shift-invariant and attractive spin-systems are **Stochastic Ising Model, Voter Model, Contact Process**.

Now we shall talk about voter model on finite graphs.

2 What Is Voter Model?

A Voter Model on a graph $G = (V, E)$ is a IPS with state-space $\{0, 1\}^V$ and flip-rates $c(x, \eta) = \sum_{y \in V | (x, y) \in E} \mathbb{1}_{\eta(y) \neq \eta(x)}$.

2.1 Interpretation Of Voter Model:

We interpret the model as follows: Suppose vertices represent individuals and the set $A = \{0, 1\}$ represents 2 political preferences, say republican (denoting by 1) vs democrat (denoting by 0). Individuals chooses a random neighbour at rate 1 and adopt the opinion of that neighbour. Thus this model encodes possible communications in the network, i.e; 2 individuals can discuss and change their opinion iff they are connected by an edge and describes the evolution of the political preferences in a given population.

2.2 Voter Model As A Markov Process:

A set V of n agents with opinions 0 or 1 are present. Independent Poisson processes with rates $q_{vw}, v \neq w$, is associated with each ordered pair of distinct agents (v, w) and $q_{vw} = 0$ if $(v, w) \notin E$. Set $q_{vv} = -\sum_{w \neq v} q_{vw}$ such that $Q = \{q_{vw} | v, w \in V\}$ be the rate matrix of a Markov Process. Here, $q_{vw} = \frac{1}{\deg(v)} \forall (v, w) \in E$ is a common choice as the model is on a graph. Observe that, $\pi_v = \frac{\deg(v)}{2|E|}$ satisfies the detailed balance equation. Hence, $\pi_v = \frac{\deg(v)}{2|E|}$ is a stationary distribution here. At the time of increment, v copies the opinion of w .

Consider $X_v(t)$ as the opinion of agent v at time t . The vector $\mathbf{X}(t) = \{X_v(t) | v \in V\}$ denotes the configuration of opinions at time t . Then $\mathbf{X}(t), t \geq 0$, is a Markov Process on $\{0, 1\}^V$ with transition rates-

$$\begin{aligned} \eta(\mathbf{x}, \mathbf{x} + e_v) &= \mathbb{1}_{X_v=0} \sum_{w | X_w=1} q_{vw} \\ \eta(\mathbf{x}, \mathbf{x} - e_v) &= \mathbb{1}_{X_v=1} \sum_{w | X_w=0} q_{vw}, \end{aligned}$$

where e_v is the unit vector with 1 at v -th co-ordinate.

2.3 Irreducibility:

When the Voter Model is on a connected (finite) graph, then we call it is irreducible as connectivity assures we travel from any site to another site in some no. of steps and hence from any configuration (other than absorbing states), we can travel through vertices and have a positive probability to get a desired configuration after travelling through some no. of steps. So, our model is **irreducible** if it is on a connected (finite) graph. Suppose we take voter model on a disconnected graph like in the figure-1.

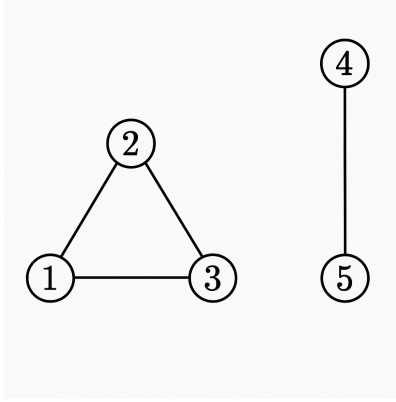


Figure 1: Disconnected graph on 5 vertices

Now assign the initial configuration of voter model on this with assigning 1 at vertices 1, 3 and 0 at vertices 2, 4, 5 as in figure-2.

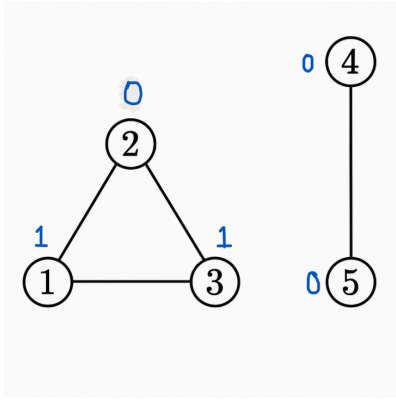


Figure 2: initial configuration

Observe that this isn't an absorbing state but we can't move to configuration 0, 0, 0, 1, 0 starting from this here, so the chain can't move to any other configuration here. So, on disconnected graph it isn't irreducible.

2.4 Aperiodicity:

The model is aperiodic clearly, because if we choose a vertex which has same state at all it's neighbours as itself, then the configuration remains same at next step, so, it is aperiodic.

2.5 Absorbing Configurations:

Notice that if the model reaches the time when all sites are at state 1 or all sites are at state 0, then we can't move to any other configurations with probability 1 (as transition rate is 0 at those cases), so these are absorbing configurations.

From Any other configurations there exist a positive probability to move to other configurations, so only these 2 configurations are absorbing.

2.6 Consensus Time:

Consensus time for voter model is the time when either all sites are at state 1 or all states are at state 0, i.e; when the system reaches to absorbing configuration and hence no transition can change the configuration further. For voter model on connected (finite) graph, consensus is guaranteed in finite time due to irreducibility of the Markov process (as we can reach any state including absorbing states starting from any non-absorbing states with a positive probability), next we shall look into the rigorous proof of this using martingales.

2.7 What Can We Say About The Probability Of Reaching Consensus On Different Absorbing States Using Optimal Stopping Time Theorem?

Let T be the random time to reach the consensus, i.e; $T = \inf\{t \geq 0 | \mathbf{X}(t) = \mathbf{0} \text{ or } \mathbf{1}\}$, where $\mathbf{0}$ denotes the configuration where all sites are 0 and $\mathbf{1}$ denotes the configuration where all sites are 1. By definition of T , $\mathbf{X}(T) = \mathbf{0}$ or $\mathbf{1}$. T is a stopping time clearly as $\{T \leq t\}$ (which means that the process reach $\mathbf{0}$ or $\mathbf{1}$ before time t) depends only on the history upto time t .

We need to show: T is finite with probability 1 if Q is irreducible, i.e; if G is **connected**.

To prove this take, $\alpha = \mathbb{P}(\mathbf{X}(T) = \mathbf{1})$. Now by A.1.5, $M(t) = \vec{\pi} \cdot \mathbf{X}(t)$ is a martingale. By optimal stopping time theorem of martingales, mentioned in A.1.4, $\mathbb{E}(M(T)) = M(0)$ which implies $\mathbb{E}(\vec{\pi} \cdot \mathbf{X}(T)) = \vec{\pi} \cdot \mathbf{X}(0)$. Now as $\mathbf{X}(T) = \mathbf{0}$ or $\mathbf{1}$,

$$\begin{aligned} \mathbb{E}(\vec{\pi} \cdot \mathbf{X}(T)) &= \sum_v 1 \cdot \pi_v \cdot P(\mathbf{X}(T) = \mathbf{1}) + \sum_v 0 \cdot \pi_v \cdot P(\mathbf{X}(T) = \mathbf{0}) \\ &= \sum_v \pi_v P(\mathbf{X}(T) = \mathbf{1}) \text{ (As 2nd term cancels out)} \\ &= P(\mathbf{X}(T) = \mathbf{1}) \text{ (As } \vec{\pi} \text{ stationary distribution)} \\ &= \alpha. \end{aligned}$$

So, $\vec{\pi} \cdot \mathbf{X}(0) = \alpha = \mathbb{P}(\mathbf{X}(T) = \mathbf{1})$ and $\mathbb{P}(\mathbf{X}(T) = \mathbf{0}) = 1 - \vec{\pi} \cdot \mathbf{X}(0)$ (as $\mathbf{X}(T) = \mathbf{0}$ or $\mathbf{1}$, which implies with probability 1 we reach consensus, i.e; $P(T < \infty) = 1$).

3 Voter Model On Connected Graph:

Already we noticed that for voter model on a connected graph, we have a finite consensus time. Now we shall introduce an upper bound on this consensus time using a dual-process which is a coalescing random walks.

3.1 Continuous Time Markov Chains and Graphical Representation:

Duality: For two Markov processes $X = (X_t)_{t \geq 0}$ and $Y = (Y_t)_{t \geq 0}$ on state-spaces $\mathcal{X}$ and $\mathcal{Y}$ respectively, we say X and Y are dual to each other with respect to duality function H , if for a bounded measurable function $H : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$, $\mathbb{E}_x(H(X_t, y)) = \mathbb{E}_y(H(Y_t, x))$, $\forall (x, y) \in (\mathcal{X} \times \mathcal{Y})$, $t \geq 0$.

Let $(Y_t)_{t \geq 0}$ denote the dual Markov process with respect to voter model $(X_t)_{t \geq 0}$ on state-space $\mathcal{X} = \{0, 1\}^{[n]}$. Since it is a spin-flip model we take state-space of dual as $\mathcal{Y} = \{A | A \subseteq [n]\}$. So $(Y_t)_{t \geq 0}$ can be interpreted as the time evolution of a collection of independent continuous time random walk that coalesce whenever two of them occupy the same site. This is known as coalescing random walk. Duality function $H(\eta, A) = \mathbb{1}_{\eta(x)=1, \forall x \in A}$, $\forall \eta \in \mathcal{X}$, $A \in \mathcal{Y}$ works here as we are interested in consensus, i.e; configurations η where $\eta(x) = \eta(y)$, $\forall x, y \in [n]$. The transition rates of $(X_t)_{t \geq 0}$ for transitions $A \rightarrow B$, where $A, B \in \mathcal{Y}$, are given by $Q(A, B) = \sum_{x \in A} \sum_{y \in [n] \setminus (A \setminus \{x\}) \cup \{y\} = B} q(x, y)$. This has the following interpretation: each $x \in A$ is removed from A at rate 1 and replaced by y with probability $q(x, y)$. Moreover, if an attempt is made to place a point at a site already occupied, then the two points coalesce. In this system of coalescing random walks, each walk independently waits an exponential time with rate 1, then moves according to $q(x, y)$, and coalesces upon landing on an occupied site.

3.2 Graphical Representation:

Consider a graph $\{(j, t) | j \in [n], t \geq 0\}$ and independent Poisson process $(N_i(t))_{t \geq 0}$ on each vertex $i \in [n]$, with rate 1 and the dynamics is as follows: If $\bar{t}$ is a time when an event occurs in Poisson process N_i , then draw an arrow from $(\bar{t}, j)$ to $(\bar{t}, i)$ where j is chosen with probability $q(i, j)$. An event represented by an arrow $j \rightarrow i$ means that at time $\bar{t}$, the voter at site i decide to copy the opinion of the voter at site j . Given any initial configuration $\eta_0 \in \{0, 1\}^{[n]}$, we let the opinions flow upwards starting from time $t = 0$ and any time they encounter the base of an arrow, they follow it's direction changing to the opinion that is at the tip of the arrow. Fix a time $t_0 > 0$ and position of a walk in (i, t_0) , $\forall i \in [n]$. We let these walks evolve independently as follows: they move downwards through the graph $\{(j, (t_0 - t)) | j \in [n], t \in [0, t_0]\}$ and any time they encounter the tip of an arrow, they follow it in the opposite direction. Furthermore, if one of them moves to a site already occupied by another walk, then the two walks coalesce into a single one.

Alternatively, we can describe the process as follows: each of the walks waits an exponential time with parameter 1 and from given current position i , it moves to j with probability $q(i, j)$ and any time when at least two of the walks meet at same site, they coalesce into single particle and continues to perform a random walk. Suppose by time t , all ancestral lineages (starting from each vertex at time t and traced back to time 0) have merged to single ancestor then the system is in consensus before or at time t . Now let $Y(t)$ be no. of particles in the system at time t and T be the random time to consensus, then $P(T > t) \leq P(Y(t) > 1)$. Alternatively, if $\tilde{T} = \inf\{t \geq 0 | Y(t) = 1\}$,

then T is stochastically dominated by $\tilde{T}$. Now, I will give an example to show how this process works.

3.3 Example:

I shall give an example using a complete graph with 5 vertices, where vertices 1,4 has opinion 0 and vertices 2,3,5 has opinion 1. Now in the below arrow-diagram in figure-3, it shows how the vertices copied opinion of it's neighbour in a given time-interval. The vertex from which an arrow starts, copies the opinion of the vertex where that arrow points.

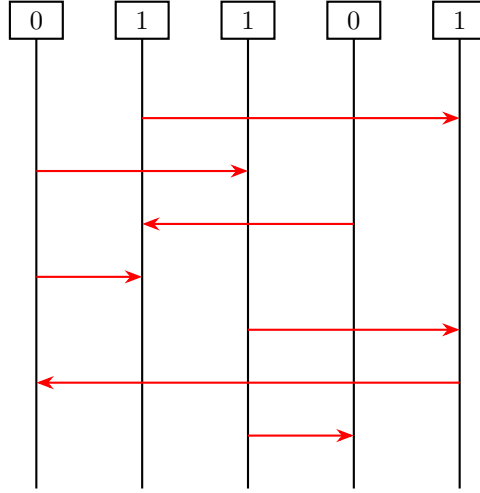


Figure 3: Here observe, at first 2 copies opinion of 5, nothing changes as both have same opinion. Then 1 copies opinion of 3, so opinion of 1 changes 1 from 0. After that 4 copies opinion of 2 and hence opinion of 4 becomes 1 from 0 and at this time-point we reach the consensus as all vertices has opinion 1.

Now we shall look into the events backwards in time. We put a particle of colour blue at each vertex at the bottom of the time-line in figure-4. (think it indicates that all vertices are of opinion 1)

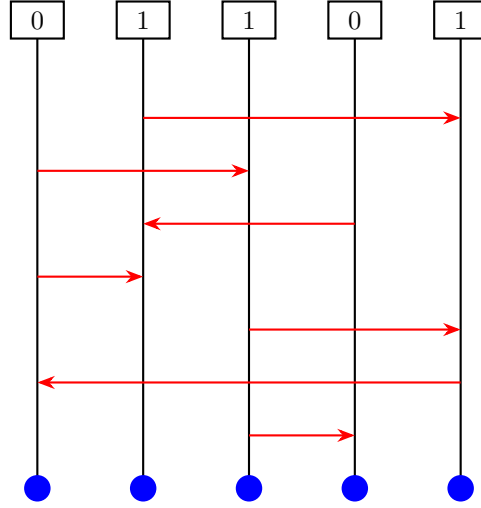


Figure 4: Putting a particle of colour blue at each vertex at the end of time-line

In backward direction of time, 3 copied opinion of 4 firstly. If we want to know about opinion of vertex 3 at end of this time-period, then what happened to vertex 3 before this time is irrelevant as it copied opinion of 4. So what happened to 4 before this time, that matters. Hence, we need to keep track, what happened to vertex 4 and by doing so, we will also see, what happened to vertex 3. So, we need to keep track of what happened to vertex 4 and by doing so, we will also see, what happened to vertex 3. So going backwards in time, we move the particle in 3 to 4 and merge with the particle in 4 because the ends of 3 and 4 are tied together. Now, we mark at particle at vertex 3 with red in figure-5 which signifies that we will not consider this particle before this time.

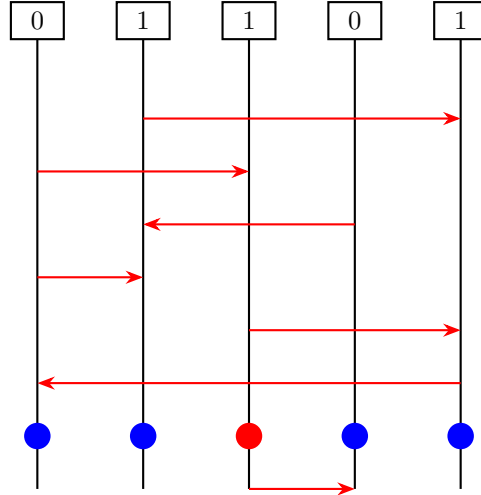


Figure 5: The red particle in vertex 3 is consider at dead and we shall not consider this particle no more in our backward-time walk.

Now, previously 5 copied the opinion of 1, hence we update our mechanism as figure-6.

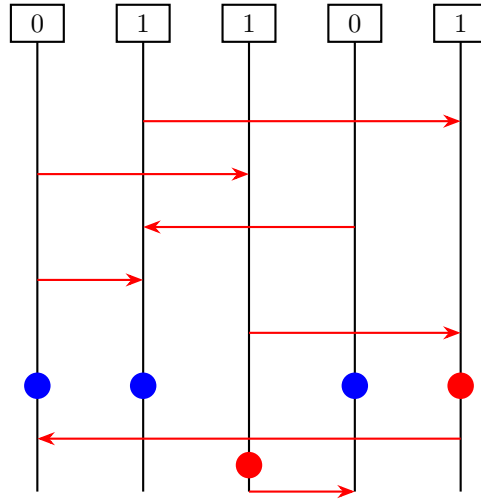


Figure 6: Now also the particle in vertex 5 is consider at dead and we shall not consider this particle no more in our backward-time walk.

Observe previously, 3 copied the opinion of 5, as we don't care about particle at 3 anymore, nothing much interesting happens, and we update our mechanism as figure-7.

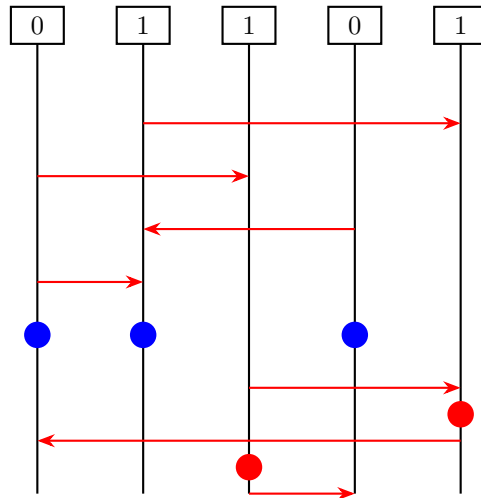


Figure 7: Particle at vertex 3 was previously dead, so nothing much interesting happen.

Now, previously 1 copied the opinion of 2, hence we update our mechanism as figure-8.

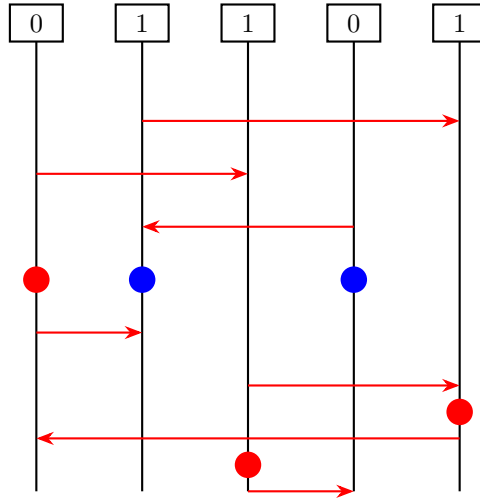


Figure 8: Now also the particle in vertex 1 is consider at dead and we shall not consider this particle no more in our backward-time walk.

Now, previously 4 copied the opinion of 2, hence we update our mechanism as figure-9.

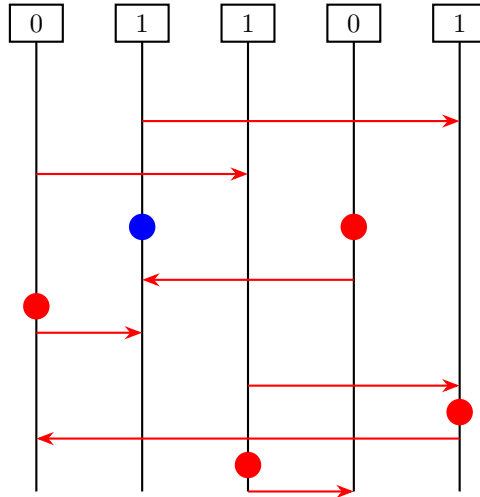


Figure 9: Now also the particle in vertex 4 is consider at dead and we shall not consider this particle no more in our backward-time walk.

Observe previously, 1 copied the opinion of 3, as we don't care about particle at 1 anymore, nothing much interesting happens, and we update our mechanism as figure-10.

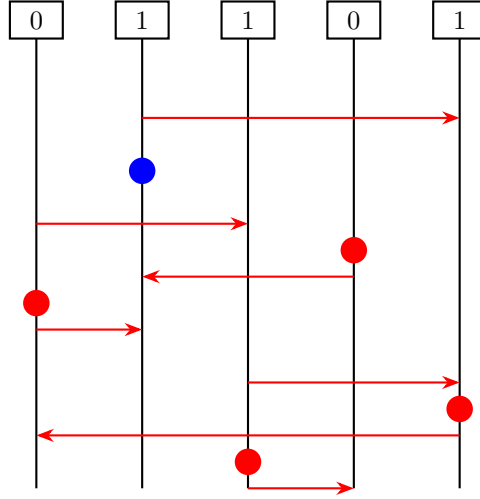


Figure 10: Particle at vertex 1 was previously dead, so nothing much interesting happen.

Previously, 2 copied the opinion of 5, so before this time opinion of 2 is irrelevant and hence we colour it as red and the particle at 2 goes to 5 and should merge with the particle at vertex 5, but there is no particle at 5 to merge with and hence the particle at 2 just moves to 5 and we update our mechanism as figure-11.

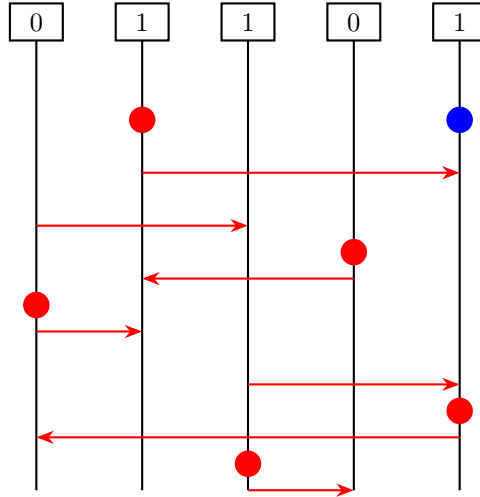


Figure 11: Now also the particle in vertex 2 is consider at dead and we shall not consider this particle no more in our backward-time walk and the particle moves to 5.

Now we can observe the particle at vertex 5 in figure-11 is the only particle left and we have reached the origin of time. This backward- time mechanism now basically says that all the vertices present

The diagram shows a quantum circuit with five qubits. The initial state of each qubit is indicated by a box above the line: 0, 1, 1, 0, and 1. The circuit consists of several gates represented by horizontal red arrows. Blue dots indicate control points, and red dots indicate target points. The gates are as follows:

- Qubit 1 (control, blue dot) to Qubit 5 (target, red dot).
- Qubit 2 (control, blue dot) to Qubit 3 (target, red dot).
- Qubit 4 (control, red dot) to Qubit 2 (target, blue dot).
- Qubit 4 (control, blue dot) to Qubit 1 (target, red dot).
- Qubit 1 (control, blue dot) to Qubit 5 (target, red dot).
- Qubit 3 (control, blue dot) to Qubit 5 (target, red dot).
- Qubit 3 (control, red dot) to Qubit 4 (target, blue dot).
- Qubit 3 (control, blue dot) to Qubit 4 (target, red dot).

3.4 Why Isn't Consensus Guaranteed Until Only One Vertex Is Left?

12

4 Some Previous Researches On Consensus Time:

Voter Model is a well-known example of opinion dynamics introduced independently in the mathematical literature by Clifford and Sudbury in the paper [4] and by Holley and Liggett in the paper [13].

Phase transition for voter model is governed by the dimension d of lattice. For $d \leq 2$ consensus occurs while for $d \geq 3$ coexistence emerges. Holley and Liggett in the paper [13] proved the presence of a continuous family of invariant measures in the supercritical regime with an index depending on the initialization of the opinions while in subcritical regime every invariant measure reduces to a convex combination of Dirac measures at fixation states. This phenomena is closely related to the recurrence and transience properties of the simple random walk on $\mathbb{Z}^d$. From the paper [8] by Cox and Greven in 1990, we get to know the following result: Suppose G_d is the average number of visits to 0 of simple random walk on $\mathbb{Z}^d$, where $\Lambda_N = [0, N)^d \cap \mathbb{Z}^d$ is a finite torus on $\mathbb{Z}^d$ with mean $\theta \in (0, 1)$ and $O_{T_N}^N$ is the fraction of opinion 1 at time T_N and δ_i indicates the distributions degenerate at i , i.e; distribution of consensus states. Then-

- (1) for $d = 1, 2$ and any $T_N \rightarrow \infty, \lim_{N \rightarrow \infty} \mathcal{L}(O_{T_N}^N) = (1 - \theta) \delta_0 + \theta \delta_1$.
- (2) for $d \geq 3$, $\lim_{N \rightarrow \infty} \mathcal{L}(O_{s\nu_N}^N) = Z_s$, $Z_0 = \theta$,

where $\nu_N = |\Lambda_N|$ and $(Z_s)_{s \geq 0}$ is the Wright-Fisher diffusion on $[0, 1]$ with diffusion constant $\frac{1}{G_d}$. The papers [9] in 1995 and [11] in 1998 by Cox, Greven and Shiga also tells that the behaviour of consensus on Λ_N is different from that on $\mathbb{Z}^d$. In particular, there is an N -dependent characteristic time scale ν_N on which the process notices that Λ_N differs from $\mathbb{Z}^d$, resulting in different behaviour for short, moderate, and long times.

For voter model on disconnected graph G , consensus time is defined as the first time when all connected components has reached their local consensus which differs across components. The order of consensus time is determined by time to reach local consensus within the largest connected component upto a constant factor.

Numerous works have aimed to establish sharp bounds on the expected consensus time. Most of them are based on the assumption that the graph is undirected which ensures the reversibility of the associated random walk and enable the application of a variety of well-established techniques. Suppose λ_2 is the 2nd largest eigenvalue of the transition matrix and ν is the ratio of 2nd moment of degree sequence and square of its 1st moment. A result in the paper [5] by Cooper, Elsasser, Ono and Radzik in 2013 applies spectral methods to determine the expected consensus time for voter model on connected graphs as $\mathbb{E}(\tau_{cons}) = \mathcal{O}(\frac{1}{(1-\lambda_2)}(\log^4 n + \frac{n}{\nu}))$. Hassin and Peleg in the paper [12] in 1999 proved that the dynamics reaches consensus on a state η_0 with probability proportional to the volume of vertices initially in state η_0 in $\mathcal{O}(n^3 \log n)$ time regardless of graph structure. Recently, in 2019 this upper bound is improved to a tight upper bound of $\mathcal{O}(n^3)$. In 1989 Cox showed for voter model on d -dimensional tori, the expected consensus time is $\mathcal{O}(n^2)$ for $d = 1$, $\mathcal{O}(n \log n)$ for $d = 2$ and $\mathcal{O}(n)$ for $d \geq 3$ in the paper [7]. This phenomena is also related to recurrence and transience of random walk. The paper also tells that for voter model on torus even if the system is on finite state-space, we assume it is on $\mathbb{Z}^d$ lattice. In 2004, in the paper [6], Cooper and Frieze studied consensus time for voter model on d -regular random graph G . They mentioned that, with high probability (i.e; probability $\rightarrow 1$) w.r.t. randomness of the environment, $\forall \epsilon > 0, \mathbb{P}_G(|\mathbb{E}(\tau_{cons}) - \frac{d-1}{d-2}n| > \epsilon) \rightarrow 0$.

In 2008 in the book [10] Durrett showed that for voter model on complete graph, the fraction of any opinions with time-scale n (no. of vertices) converges in distribution to Wright-Fisher Diffusion $(W_t)_{t \geq 0}$ where W_t is the solution to 1-dimensional Stochastic Differential Equation $dW_t = \sqrt{2W_t(1-W_t)}dB_t$ where $(B_t)_{t \geq 0}$ is the standard Brownian motion for sufficiently large n . After that in 2016 the paper [3], it is shown that for voter model on static random d -regular graph the fraction of any opinions with time-scale n (no. of vertices) converges in distribution to Wright-Fisher Diffusion $(W_t)_{t \geq 0}$ where W_t is the solution to 1-dimensional Stochastic Differential Equation $dW_t = \sqrt{2\frac{(d-2)}{(d-1)}W_t(1-W_t)}dB_t$ where $(B_t)_{t \geq 0}$ is the standard Brownian motion for sufficiently large n . In the paper [2] it is shown that for voter model on random d -regular graphs with rewiring dynamics (which means pairs of edges can be broken into four half-edges and subsequently reconnected at random) the fraction of either one of the opinions with time-scale n (no. of vertices) converges to Wright-Fisher Diffusion $(W_t)_{t \geq 0}$ where W_t is the solution to 1-dimensional Stochastic Differential Equation $dW_t = \sqrt{2\nu W_t(1-W_t)}dB_t$ where $(B_t)_{t \geq 0}$ is the standard Brownian motion and ν is the diffusion constant depended on d and v where each edge is rewired in $\exp(v)$ rate, for sufficiently large n .

For voter model on complete graph the no. of discordant edges is the product of no. of vertices with 2 respective opinions, hence the fraction of discordant edges (of voter model on complete graph) converges in distribution to the process $(W_t(1-W_t))_{t \geq 0}$ for sufficiently large no. of vertices. But it isn't the case for voter models on other geometries. In the paper [1] it is shown that for voter model on random d -regular graph, if initially 2 opinions are drawn independently, from a Bernoulli(u) distribution with $u \in (0, 1)$, then the fraction of discordant edges is decreasing and stabilises to a value that depends on d and u . They also showed that on time-scale n , the fraction of discordant edges behaves as a deterministic function of Wright-Fisher diffusion $(W_t)_{t \geq 0}$ and converges in distribution to $2\frac{(d-2)}{(d-1)}W_{2t\frac{d-2}{d-1}}(1-W_{2t\frac{d-2}{d-1}})$.

We shall mainly focus on finding this consensus time for voter model on complete graph and introduce a process (using which we can give it an upper bound for voter model on connected graph).

5 Voter Model On Complete Graph:

Consider a complete graph on n vertices. Set $Y(t)$ = no. of nodes in state 1 at time t .

Theorem 1. $Y(t)$ is a martingale for $t \in \mathbb{R}^+$.

Proof. To prove this, we look at the change in expectation over infinitesimal time-intervals.

As it is a complete graph, with probability $\frac{1}{n}$, we can travel from any vertex to any other vertex. Suppose $Y(t) = k$, i.e; at time t , we have k vertices with opinion 1. Then transition rate is:

$$p_{k,k+1} = \frac{k(n-k)}{n} \forall k = 0, \dots, n-1$$

$$p_{k,k-1} = \frac{(n-k)k}{n} \forall k = 1, \dots, n$$

At each step, either we have a (+1) transition or a (-1) transition in any small time-interval δ . So,

$$\begin{aligned} \mathbb{E}(Y(t+\delta) - Y(t) | Y(t) = k, Y(s), 0 \leq s \leq t) &= \frac{k(n-k)\delta}{n} + o(\delta) - \left(\frac{k(n-k)\delta}{n} + o(\delta) \right) \\ &= o(\delta), \quad \forall t \in \mathbb{R}^+, \quad \forall k \in \{0, 1, \dots, n\} \end{aligned}$$

So, $Y(t)$ is a martingale for $t \in \mathbb{R}^+$. □

Theorem 2. The proportion of nodes at time t with opinion 1 is $f(t) = \frac{Y(t)}{n}$ is a martingale for $t \in \mathbb{R}^+$.

Proof. To prove this, we look at the change in expectation over infinitesimal time-intervals.

As it is a complete graph, with probability $\frac{1}{n}$, we can travel from any vertex to any other vertex. Now $f(t) \in \{0, \frac{1}{n}, \frac{2}{n}, \dots, 1\}$. Suppose $f(t) = \frac{k}{n}$, i.e; at time t , we have k vertices with opinion 1. Then transition rate is

$$p_{k,k+1} = \frac{k(n-k)}{n} \quad \forall k = 0, \dots, n-1$$

$$p_{k,k-1} = \frac{(n-k)k}{n} \quad \forall k = 1, \dots, n$$

At each step, either we have a (+1) transition or a (-1) transition in any small time-interval δ , i.e; either we have a $\frac{(+1)}{n}$ increase or a $\frac{(-1)}{n}$ decrease in proportion of nodes with opinion 1. So,

$$\begin{aligned} \mathbb{E}(Y(t+\delta) - Y(t) | Y(t) = k, Y(s), 0 \leq s \leq t) &= \frac{1}{n} \cdot \frac{k(n-k)\delta}{n} + o(\delta) - \frac{1}{n} \left(\frac{k(n-k)\delta}{n} + o(\delta) \right) \\ &= o(\delta), \quad \forall t \in \mathbb{R}^+, \quad \forall k \in \{0, 1, \dots, n\}. \end{aligned}$$

So, $f(t)$ is a martingale for $t \in \mathbb{R}^+$. □

Theorem 3. *What is the probability that voter model on complete graph reaches all 1 configuration with probability $\frac{k}{n}$.*

Proof. For voter model on complete graph, let T be the random time to hit an absorbing state. Then $T = \inf\{t \geq 0 | Y(t) = 0 \text{ or } n\}$. T is a stopping time clearly. Also, $Y(t)$ is bounded between 0 and n , and T is finite almost surely (as it is the time to hit an absorbing state in a finite state Markov Chain where the remaining states form a transient communicating class.) All the conditions of Optimal Stopping Theorem are satisfied. So, $\mathbb{E}(Y(T)) = Y(0)$, but $\mathbb{E}(Y(T)) = n\mathbb{P}(Y(T) = n) + 0\mathbb{P}(Y(T) = 0) = n\mathbb{P}(Y(T) = n)$. So if $Y(0) = k$, then $\mathbb{P}(Y(T) = n) = \frac{k}{n}$. □

Theorem 4. *Exact Consensus time in complete graph is $(n - 1)$.*

Proof. For complete graph on n vertices, $q_{vw} = \frac{1}{n-1}, \forall v \neq w$. In the coalescing random walk model on complete graph, due to symmetry, it doesn't matter where exactly the particles are. It is enough to keep track of only no. of particles because the future evolution looks the same no matter where the particles are allocated. So for complete graph, the no. of particles evolve a Markov process. Denote $\tau_k = \inf\{t \geq 0 | Y(t) = k\}$ to be the first time that there are exactly k particles left in the coalescing random walk process. So, $\tau_n = 0$ and we want to evaluate τ_1 to get the consensus time.

Suppose there is a particle at vertex v and v talks to another vertex w . If there was no particle at w , nothing changes, i.e; the no. of particles remains same. If there was a particle at vertex w , these two particles merges to one particle decreasing no. of particles by 1. So, we need to look at all pairs of occupied vertices and the rate communication happens between them (which is $\frac{1}{n-1}$ here). For k particles, $\binom{k}{2}$ pairs of occupied vertices exist and there is 2 possible direction for each edge. So, there are $k(k-1)$ directed edges in total through which 2 particles can merge. Now this moves happen at times of independent Poisson process of rate $\frac{1}{n-1}$. So, times between moves follows exponential distribution (by properties of Poisson Process A.3.1) and the time is the minimum of $k(k-1)$ many independent $\exp(\frac{1}{n-1})$ random variables. So, by A.2.1, the times between moves follows $\exp(\frac{k(k-1)}{n-1})$ distribution. Hence, the event $(\tau_{k-1} - \tau_k)$ follows $\exp(\frac{k(k-1)}{n-1})$ distribution and by A.2.2, $\mathbb{E}(\tau_{k-1} - \tau_k) = \frac{n-1}{k(k-1)}$.

Now we want to find $\mathbb{E}(\tau_1)$.

$$\begin{aligned} \mathbb{E}(\tau_1) &= \sum_{k=2}^n \mathbb{E}(\tau_{k-1} - \tau_k) \\ &= \sum_{k=2}^n \frac{n-1}{k(k-1)} \\ &= \sum_{k=2}^n \left(\frac{n-1}{(k-1)} - \frac{n-1}{k} \right) \\ &= (n-1). \end{aligned} \tag{1}$$

Hence, we proved that expected consensus time is $(n - 1)$ in this case. □

5.1 A Simulation Result and R-code For Consensus Time of Voter Model On Complete Graph with 100 Vertices:

In the following, the R-code for simulating a voter model on a complete graph with Bernoulli(0.5) clock on each vertices is given and also the R-code and plots for fractions of 1 and 0 opinions with respect to time is given. The fraction of any opinions (of voter model on complete graph) with time-scale n (no. of vertices) converges in distribution to Wright-Fisher Diffusion $(W_t)_{t \geq 0}$ where W_t is the solution to 1-dimensional Stochastic Differential Equation $dW_t = \sqrt{2W_t(1 - W_t)}dB_t$ where $(B_t)_{t \geq 0}$ is the standard Brownian motion.

We also simulate fraction of discordant edges. Discordant edges indicates those edges which connects vertices with two different opinions, i.e; $e = (x, y) \in E$ such that $\eta(x) \neq \eta(y)$ is called discordant edge. So after reaching consensus no. of discordant edges become 0 as all vertices has same opinion after reaching consensus. The fraction of discordant edges (of voter model on complete graph) converges in distribution to the process $(W_t(1 - W_t))_{t \geq 0}$ for sufficiently large no. of vertices.

Observe in the following plots that as time increases, the fraction of one of the opinions tends to 0 and another one tends to 1 which indicates eventually we reach consensus, and also the fraction of discordant edges tends to 0 exponentially.

5.1.1 Pseudocode:

Algorithm 1 Voter Model Simulation on a Complete Graph

Require: n : number of nodes, M : number of simulations

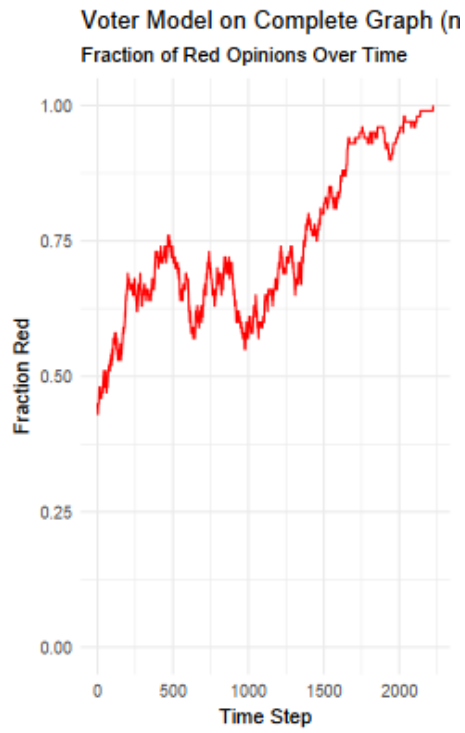
Ensure: Average consensus time, opinion dynamics plots

```
1: function VOTER MODEL SIMULATION( $n$ )
2:   Initialize opinions on  $n$  nodes: each node  $\sim$  Bernoulli(0.5)
3:   Compute initials:
   • Fraction of opinion-1 nodes
   • Fraction of opinion-0 nodes
   • Fraction of discordant edges
4:   Store values in 3 different history arrays
5:    $t \leftarrow 0$ 
6:   while more than one opinion exists do
7:      $t \leftarrow t + 1$ 
8:     Randomly select node  $i$ 
9:     Randomly select neighbour  $j \neq i$ 
10:    Update:  $\text{opinion}(i) \leftarrow \text{opinion}(j)$ 
11:    Recompute and store fractions (blue, red, discordant)
12:  end while
13:  return consensus time  $t$ , 3 history arrays
14: end function
```

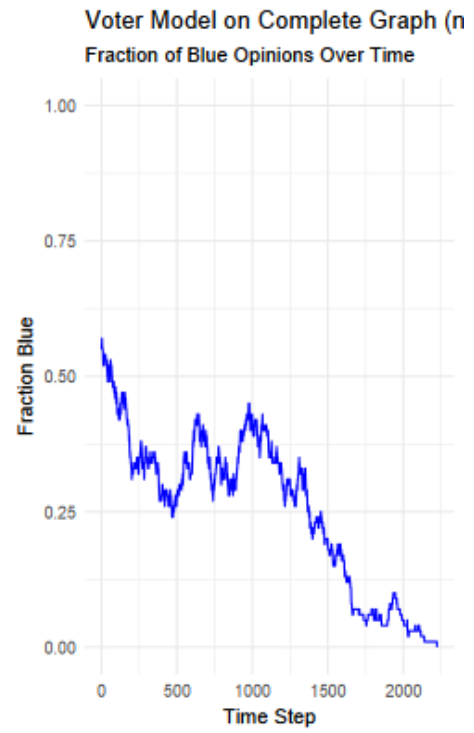
15: **Simulating The Main Program:**

```
16: Set parameters  $n, M$ 
17: for  $m = 1$  to  $M$  do
18:   Run VOTER MODEL SIMULATION( $n$ )
19:   Record consensus time of  $m$ th simulation
20: end for
21: Compute average consensus time over  $M$  runs
22: Plot for one run:
   • Fraction of opinion-1 nodes vs. time
   • Fraction of opinion-0 nodes vs. time
   • Fraction of discordant edges vs. time
```

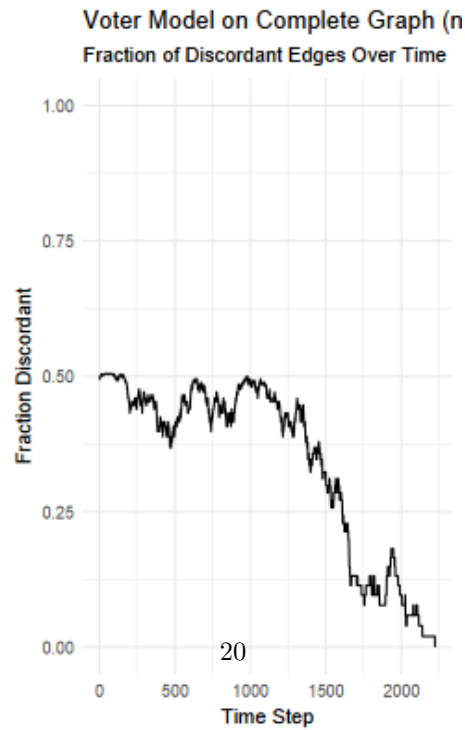
5.1.2 R-output of the Plots:



(a) Fraction of vertices with opinion-0



(b) Fraction of vertices with opinion-1



(c) Fraction of discordant edges (edges connecting a 0-opinion with a 1-opinion)

Figure 13: Voter Model Simulation on Complete Graph

A Appendix

A.1 Martingales and Optimal Stopping Theorem:

A.1.1 Discrete-Time Martingales:

Let $X_t, \forall t \in \mathbb{N}$ be a real-valued stochastic process. We say X_t is a martingale if $\forall t \in \mathbb{N}, \mathbb{E}(X_{t+1} | \text{history upto time } t) = X_t$. Applying tower property in the definition of martingale, we can say for martingales, $\mathbb{E}(X_t) = X_0, \forall t \in \mathbb{N}$. This property has nothing to do with whether or not X_t is a Markov Chain.

Note: Intuitively, a martingale is like one's fortune in a fair game of chance and you are allowed to adopt an arbitrarily complicated strategy to decide how much to gamble at each step.

A.1.2 Continuous-Time Martingales:

Let $X_t, \forall t \in \mathbb{R}$ is real-valued stochastic process. We say X_t is a martingale if $\mathbb{E}(X_t | \text{history upto time } s) = X_s, \forall s \in [0, t]$. Then for all fixed t again by tower property, we get $\mathbb{E}(X_t) = X_0$.

A.1.3 Stopping Time:

Definition 1. A random time T is called a stopping time if for any given $t \in \mathbb{N}$, the event $\{T \leq t\}$ is determined by the history upto time t .

We can give a more precise definition using filtrations $\{\mathcal{F}_t\}_{t \in \mathbb{R}}$ which is an increasing sequence (i.e; $\mathcal{F}_s \subseteq \mathcal{F}_t, \forall t \geq s$) of σ -algebras. We call an event $\mathcal{F}_t$ -measurable if the event does not depend on $\mathcal{F}_r, r > t$, i.e; if the event depends only the history upto time t .

Definition 2. A random time T is called a stopping time w.r.t. a filtration $\mathcal{F}_t, \forall t$ if the event $\{T \leq t\}, \forall t$ is $\mathcal{F}_t$ -measurable.

A.1.4 Optimal Stopping Time Theorem:

Question: For both discrete and continuous time Martingales, we saw for all fixed $t, \mathbb{E}(X_t) = X_0$ holds, but does it also hold for random T rather than fixed t ?

For both discrete and continuous time, if X_t is a **bounded martingale** which means $\exists a, b \in \mathbb{R} \ni X_t \in [a, b], \forall t$ with probability 1, then for $t = T$ (where T is **stopping time** and $P(T < \infty) = 1$) we get $\mathbb{E}(X_T) = X_0$.

Note: Boundedness is a strong assumption and there are weaker conditions on X_t under which the optimal stopping time theorem can be stated. But they require more complicated conditions which may not be completely straight-forward to check. In the applications we consider the martingale will always be bounded and the finiteness of the stopping time will follow from finiteness of the state-space.

A.1.5 An Auxiliary Markov Process and a Martingale:

Consider the random walk on V with rate-matrix Q . Here edge-set is defined as, $E = \{(v, w) \in (V \times V) | q_{vw} > 0\}$. If G is strongly connected then the random walk is an irreducible Markov Process on the state-space V .

Let $\vec{\pi}$ be the vector of unique stationary distribution of this Markov Process.

Lemma 1. *Consider $M(t) = \vec{\pi} \cdot \mathbf{X}(t)$. Then $M(t)$ is a martingale w.r.t. the filtration generated by $\{\mathbf{X}(t) | t \geq 0\}$.*

Proof. Suppose $\mathbf{X}(t) = \mathbf{x}$. Then $\mathbb{P}(\mathbf{X}(t+\delta) = \mathbf{x} + e_v) = \mathbb{1}_{X_v=0} \sum_{w|X_w=1} q_{vw} \delta + o(\delta)$ and $\mathbb{P}(\mathbf{X}(t+\delta) = \mathbf{x} - e_v) = \mathbb{1}_{X_v=1} \sum_{w|X_w=0} q_{vw} \delta + o(\delta)$. (As, we started independent Poisson processes at each pair of edges and in a small δ time interval at most 1 event takes place and now see A.3.1).

$$\text{Then } M(t+\delta) = \vec{\pi} \cdot \mathbf{X}(t+\delta) = \begin{cases} M(t) + \pi_v, & \text{if } \mathbf{X}(t+\delta) = \mathbf{x} + e_v \\ M(t) - \pi_v, & \text{if } \mathbf{X}(t+\delta) = \mathbf{x} - e_v \end{cases}$$

Hence,

$$\begin{aligned} \mathbb{E}(M(t+\delta) - M(t) | \mathbf{X}(t) = \mathbf{x}; \mathbf{X}_s, \forall s \in [0, t]) &= \pi_v \mathbb{P}(\mathbf{X}(t+\delta) = \mathbf{x} + e_v) - \pi_v \mathbb{P}(\mathbf{X}(t+\delta) = \mathbf{x} - e_v) \\ &= \pi_v (\mathbb{1}_{X_v=0} \sum_{w|X_w=1} q_{vw} \delta + o(\delta)) - \pi_v (\mathbb{1}_{X_v=1} \sum_{w|X_w=0} q_{vw} \delta \\ &\quad + o(\delta)) \\ &= \delta (\sum_{v,w} \pi_v q_{vw} (\mathbb{1}_{X_v=0, X_w=1} - \mathbb{1}_{X_v=1, X_w=0}) + o(\delta)) \\ &\quad (\text{As } \pi_v \cdot o(\delta) = o(\delta)) \\ &= \sum_{v|X_v=0} \sum_{w|X_w=1} q_{vw} \delta \pi_v - \sum_{v|X_v=1} \sum_{w|X_w=0} q_{vw} \delta \pi_v + o(\delta) \\ &= (\sum_{v|X_v=0} \sum_{w|X_w=1} q_{vw} \delta \pi_v + \sum_{v|X_v=0} \sum_{w|X_w=0} q_{vw} \delta \pi_v) \\ &\quad - (\sum_{v|X_v=0} \sum_{w|X_w=0} q_{vw} \delta \pi_v - \sum_{v|X_v=1} \sum_{w|X_w=0} q_{vw} \delta \pi_v) + o(\delta) \\ &= \sum_{v|X_v=0} \sum_w q_{vw} \delta \pi_v - \sum_{w|X_w=0} q_{vw} \sum_v \delta \pi_v + o(\delta) \\ &= 0 - 0 + o(\delta) \\ &= o(\delta). \end{aligned}$$

In (2), First term cancels out as $\sum_w q_{vw} = 0$, Q being rate-matrix and second term cancels out as $\sum_v \pi_v q_{vw} = 0$, $\vec{\pi}$ being stationary distribution. So, $M(t)$ is a martingale. $\square$

So, for voter model on connected (finite) graph $G = (V, E)$, $M(t) = \frac{\sum_v \deg(v) X_v(t)}{2|E|}$ is a martingale.

A.2 Exponential Distribution:

A.2.1 Minimum of IID Exponentials:

Theorem 5. Suppose we have m independent $\exp(\lambda)$ random variables $X_1, \dots, X_m$. Then $X_{(1)}$ follows $\exp(m\lambda)$ distribution.

Proof. By density formula of order statistics, we know $f_{X_{(1)}}(x) = mf_X(x)(1 - F_X(x))^{m-1}$. Following this formula here, $f_{X_{(1)}}(x) = m\lambda e^{-\lambda x}(1 - (1 - e^{-\lambda x}))^{m-1} = m\lambda e^{-m\lambda x}$ which means $X_{(1)}$ follows $\exp(m\lambda)$ distribution. $\square$

A.2.2 Expectation of Exponential Distribution:

Suppose X follows $\exp(\mu)$, then $\mathbb{E}(X) = \frac{1}{\mu}$.

A.3 Stochastic Process:

A.3.1 Poisson Process:

Suppose $N(t)$ is the no. of occurrences of a certain type of event in $(0, t)$. Then $N(t)$ is a Poisson process with rate λ if-

1. $N(0) = 0$
2. $N(t)$ has independent and stationary increments.
3. $P(N(t) = 1) = \lambda t + o(t)$
4. $P(N(t) \geq 2) = o(t)$.

We can show from this that $N(t)$ follows $\text{Poi}(\lambda t)$ and $\forall s \geq 0, (N(t+s) - N(s))$ follows $\text{Poi}(\lambda t)$.

Time between 2 events in Poisson Process: The elapsed time between occurrences of 2 events in Poisson process follows $\exp(\lambda)$ distribution.

(See [14] for reference).

A.3.2 Standard Brownian Motion:

A standard Brownian motion is a random process $B = \{B_t | t \geq 0\}$ with state-space $\mathbb{R}$ that satisfies-

1. $B_0 = 0$
2. It has stationary increments, i.e; for $s, t \geq 0, s < t$, the distribution of $(B_t - B_s)$ is the same as the distribution of B_{t-s} .
3. It has independent increments, i.e; for $t_1, t_2, \dots, t_n \geq 0, t_1 < t_2 < \dots < t_n$, the random variables $B_{t_1}, (B_{t_2} - B_{t_1}), \dots, (B_{t_n} - B_{t_{n-1}})$ are independent.
4. $\forall t, u \geq 0, B_{t+u} - B_t \sim \mathcal{N}(0, u)$.
5. B_t is almost surely continuous in t

Furthermore, $(B_{t_1}, B_{t_2}, \dots, B_{t_n}) \sim \mathcal{N}_n(\mathbf{0}, \Sigma)$ where $\Sigma = ((\min\{t_i, t_j\}))_{i,j}$.

A.3.3 Stochastic Differential Equation:

A stochastic differential equation (SDE) is a differential equation in which one or more of the terms is a stochastic process, resulting in a solution which is also a stochastic process. It is basically an equation of the form

$$dX_t = b(X_t, t) dt + \sigma(X_t, t) dB_t, \quad (2)$$

where

- X_t is the stochastic process which is to be evaluated
- $\mu(X_t, t)$ is the given *drift coefficient* which describes the average or expected rate of change of the stochastic process (to be evaluated) with respect to time.
- $\sigma(X_t, t)$ is the given *diffusion coefficient* which scales the size of the fluctuations caused by Brownian motion
- B_t is a standard Brownian motion (Wiener process).

B R-code:

```
1 install.packages("ggplot2")
2 library(ggplot2)
3 # Function to simulate one run of voter model
4 voter_model_simulation <- function(n) {
5   # Initialize opinions: 1 = blue, 0 = red
6   opinions <- rbinom(n, 1, 0.5)
7   fraction_blue <- mean(opinions)
8   fraction_red <- (1 - fraction_blue)
9   k <- sum(opinions) # no. of blue
10  total_edges <- n*(n-1)/2
11  frac_discordant <- k*(n-k)/total_edges # fraction of edges connecting 0 with 1
12
13  history <- c(fraction_blue)
14  history_1 <- c(fraction_red)
15  history_discordant <- c(frac_discordant)
16  time <- 0
17
18  while (length(unique(opinions)) > 1) {
19    time <- time + 1
20    i <- sample(1:n, 1) # pick a random node
21    j <- sample(setdiff(1:n, i), 1) # pick a random neighbor (different node)
22    opinions[i] <- opinions[j] # copy neighbor's opinion
23    #update
24    fraction_blue <- mean(opinions)
25    fraction_red <- (1 - fraction_blue)
26    k <- sum(opinions)
27    frac_discordant <- k*(n-k)/total_edges
28    history <- c(history, fraction_blue)
29    history_1 <- c(history_1, fraction_red)
30    history_discordant <- c(history_discordant, frac_discordant)
31  }
32
33  return(list(time = time, history = history, history_1 = history_1, history_
34             discordant = history_discordant)
35 )
```



```

35 # Parameters
36 n <- 100 # number of nodes
37 num_simulations <- 100 # for averaging consensus time
38 consensus_times <- numeric(num_simulations)
39 }
40 # Run multiple simulations to estimate average consensus time
41 for (sim in 1:num_simulations) {
42   result <- voter_model_simulation(n)
43   consensus_times[sim] <- result$time
44 }
45
46 avg_consensus_time <- mean(consensus_times)
47 cat("Average consensus time over", num_simulations, "simulations:", avg_consensus_
    time, "\n")
48
49 # Plot evolution of one run
50 set.seed(42)
51 example_run <- voter_model_simulation(n)
52 df <- data.frame(
53   step = 0:(length(example_run$history) - 1),
54   fraction_blue = example_run$history
55 )
56
57 df_1 <- data.frame(
58   step = 0:(length(example_run$history_1) - 1),
59   fraction_red = example_run$history_1
60 )
61
62 df_discordant <- data.frame(
63   step = 0:(length(example_run$history_discordant) - 1),
64   frac_discordant = example_run$history_discordant
65 )
66
67 ggplot(df, aes(x = step, y = fraction_blue)) +
68   geom_line(color = "blue") +
69   ylim(0, 1) +
70   labs(title = paste("Voter Model on Complete Graph (n =", n, ")"),
71        subtitle = "Fraction of Blue Opinions Over Time",
72        x = "Time Step",
73        y = "Fraction Blue") +
74   theme_minimal()
75
76 ggplot(df_1, aes(x = step, y = fraction_red)) +
77   geom_line(color = "red") +
78   ylim(0, 1) +
79   labs(title = paste("Voter Model on Complete Graph (n =", n, ")"),
80        subtitle = "Fraction of Red Opinions Over Time",
81        x = "Time Step",
82        y = "Fraction Red") +
83   theme_minimal()
84
85 ggplot(df_discordant, aes(x = step, y = frac_discordant)) +
86   geom_line(color = "black") +
87   ylim(0, 1) +
88   labs(title = paste("Voter Model on Complete Graph (n =", n, ")"),
89        subtitle = "Fraction of Discordant Edges Over Time",
90        x = "Time Step",
91        y = "Fraction Discordant") +

```

92

```
theme_minimal()
```

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